

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

BE-CSE CURRICULUM AND DETAILED SYLLABUS - SEMESTER 6

Academic Year 2024-2025

Course 1: CS8601 - Artificial Intelligence & Machine Learning (Credits: 4)

Course Objectives: To understand intelligent agents, search algorithms, knowledge representation, supervised learning, neural networks, and modern transformers.

- **Unit 1 - Intelligent Agents & Heuristic Search:** Agents and environments, State space search, BFS, DFS, Heuristic search, A* algorithm, AO* search, Adversarial search, Minimax with Alpha-Beta pruning.
- **Unit 2 - Knowledge Representation & Reasoning:** First-order predicate logic, Unification, Resolution refutation, Bayesian networks, Hidden Markov Models, Dempster-Shafer theory.
- **Unit 3 - Classical Machine Learning:** Supervised vs Unsupervised learning, Linear and Logistic Regression, Decision Trees (ID3, CART), Support Vector Machines (SVM), Random Forests, K-Means clustering.
- **Unit 4 - Deep Learning Foundations:** Feedforward Neural Networks, Backpropagation, CNN for Computer Vision, RNN, LSTM, and GRU for sequential data.
- **Unit 5 - Generative AI & LLMs:** Attention mechanism, Transformer architecture, Self-attention, Pre-trained language models (BERT, GPT), Prompt engineering, RAG architecture.

Prescribed Textbooks: Stuart Russell & Peter Norvig, 'Artificial Intelligence: A Modern Approach', 4th Edition; Ian Goodfellow, 'Deep Learning', MIT Press.

Course 2: CS8602 - Cloud Computing & Distributed Systems (Credits: 3)

- **Unit 1 - Cloud Architecture & Models:** NIST cloud definitions, IaaS, PaaS, SaaS, Public, Private, Hybrid clouds, Virtualization (Hypervisors Type 1 & 2), Containerization using Docker.
- **Unit 2 - Cloud Orchestration & Microservices:** Kubernetes clusters, pods, deployments, Service Mesh, RESTful microservice communication, API gateways.
- **Unit 3 - Distributed Data Processing:** Hadoop architecture, HDFS, MapReduce programming paradigm, Apache Spark distributed in-memory computations.
- **Unit 4 - Cloud Storage & Security:** Object storage (AWS S3), NoSQL stores, IAM roles, Data encryption in transit and at rest, Cloud disaster recovery.

Prescribed Textbooks: Rajkumar Buyya, 'Mastering Cloud Computing', McGraw Hill.

Course 3: CS8603 - Compiler Design (Credits: 4)

- **Unit 1 - Lexical Analysis:** Regular expressions, Finite Automata (DFA, NFA), Lex lexical analyzer generator.
- **Unit 2 - Syntax Analysis:** Context-free grammars, Top-down parsing (LL(1)), Bottom-up parsing (Shift-reduce, LR(0), SLR, CLR, LALR parsers), Yacc parser generator.
- **Unit 3 - Syntax-Directed Translation & Type Checking:** Syntax-directed definitions, S-attributed and L-attributed definitions, Intermediate code representations (Three-address code, Quadruples, Triples).
- **Unit 4 - Code Optimization & Generation:** Basic blocks and flow graphs, Loop optimization, Common subexpression elimination, Dead code elimination, Register allocation and assignment.

Prescribed Textbooks: Alfred Aho, Monica Lam, Ravi Sethi, Jeffrey Ullman, 'Compilers: Principles, Techniques, and Tools' (Dragon Book), 2nd Edition.

Course 4: CS8604 - Web Technologies & Full-Stack Development (Credits: 3)

Modern web architecture, HTML5 semantic elements, responsive CSS with Flexbox and Grid, modern ES6+ JavaScript, React component state lifecycle, Node.js and Express backend API design, MongoDB integration, JWT authentication.